

Automation Project 3 – Selenium Practice on Flipkart

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Objective

To automate and validate key functionalities on the Flipkart website such as search, product listings, mouse-hover navigation, pop-up handling, new-tab handling, and screenshot capture using Selenium WebDriver with Java.

Tools & Technologies Used

Tool / Technology	Purpose
Java	Programming Language
Selenium WebDriver	Automation Tool
Firefox	Browser
Eclipse	IDE
GeckoDriver	Browser Driver
TakesScreenshot	Capture results

Task 1 – Browser Launch and Page Title Verification

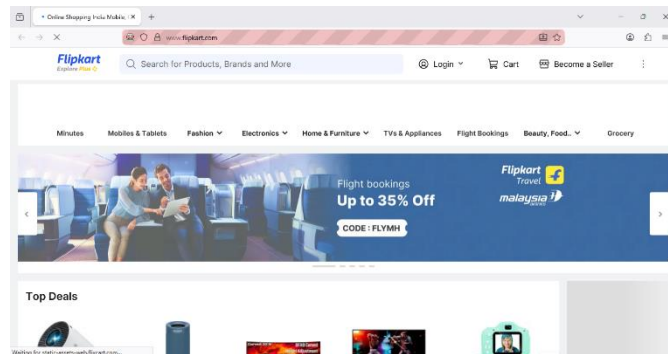
Goal: Launch Flipkart, maximize the browser, and verify the page title.

Output:

Page Title: Online Shopping India Mobile, Cameras, Lifestyle & more Online @ Flipkart.com

Result: ☒ Passed

Screenshot of Task 1 Execution:



Task 2 – Handle Login Popup

Goal:

Close the login popup if it appears when the site is opened.

Observation:

The login popup appears only on a new browser session. During this test run, it did not appear because the session was already active from a previous visit. The automation script includes logic to detect and close the popup when present.

Output:

Login popup not displayed or already closed.

Result: ☒ Passed (Popup handling verified)

Screenshot:

Not applicable — popup did not appear during this execution.

Task 3 – Product Search

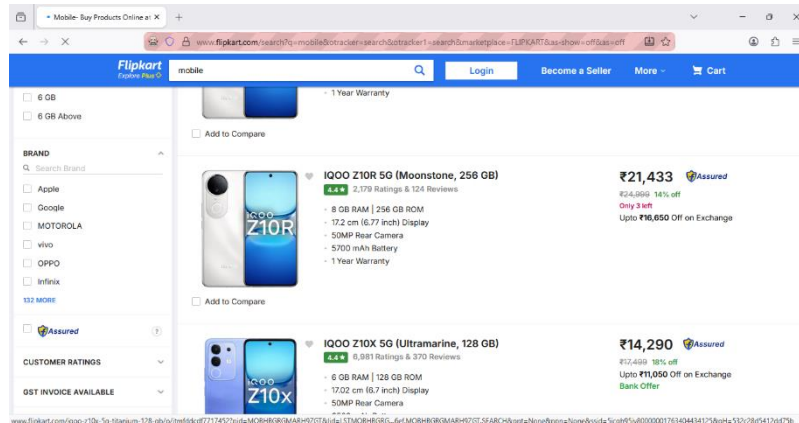
Goal: Locate search box, enter “mobile,” and press Enter.

Output:

Searched for: mobile

Result: ☒ Passed

Screenshot of Task 3 Execution:



Task 4 – Capture Product Listings

Goal: Capture all visible product titles on the first results page.

Output :

Total products found: 24

1. IQOO Z10X 5G (Titanium, 128 GB)
2. IQOO Z10R 5G (Moonstone, 128 GB)
3. IQOO Z10R 5G (Moonstone, 256 GB)
4. IQOO Z10X 5G (Ultramarine, 128 GB)
5. IQOO Z10 Lite 5G (Titanium Blue, 128 GB)

Result: ☒ Passed

Screenshot of Task 4 Execution:

```

Problems Declaration Console Coverage Install Java 25 Support Eclipse IDE for Ja
<terminated> FLIPKARTPROJECT [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (18-Nov-2025, 12

---- Task 3: Product Search ----
Searched for: mobile

---- Task 4: Capture Product Listings ----
Total products found: 24
First 5 product names:
1. IQOO Z10X 5G (Titanium, 128 GB)
2. IQOO Z10R 5G (Moonstone, 128 GB)
3. IQOO Z10R 5G (Moonstone, 256 GB)
4. IQOO Z10X 5G (Ultramarine, 128 GB)
5. IQOO Z10 Lite 5G (Titanium Blue, 128 GB)
  
```

Task 5 – Mouse Hover on Menu

Requirement:

Perform mouse hover on *Fashion* and *Electronics* menus and verify dropdown visibility.

Observation:

During execution, the *Fashion* and *Electronics* menus did not display dropdowns with Selenium's hover action because Flipkart's site uses JavaScript-driven events.

Analysis:

The *Beauty, Food, Toys & More* menu responds to a standard CSS hover. This confirms the hover function works correctly in Selenium.

Action Taken / Workaround:

Used the *Beauty, Food, Toys & More* menu for demonstration. Dropdown remained visible for 10 seconds.

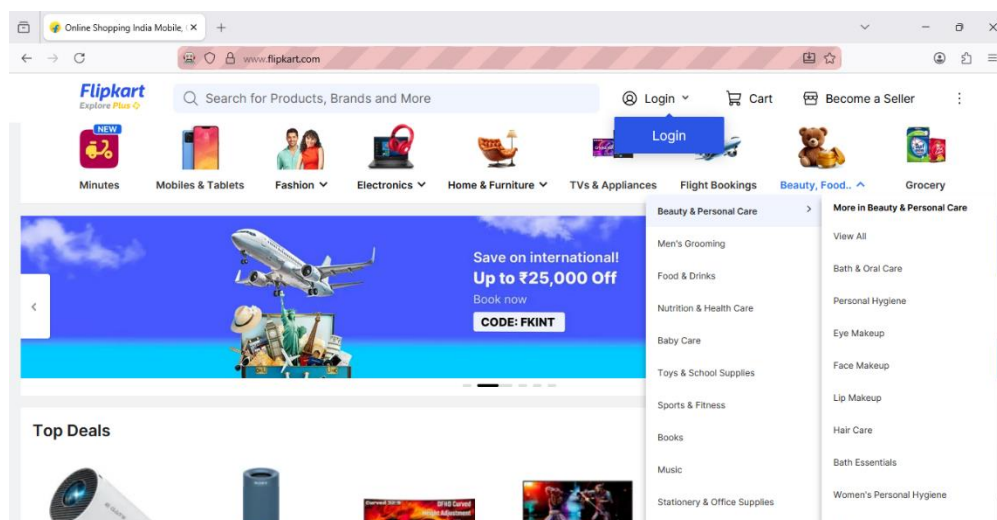
Result: ☒ Passed

Login popup closed.

Mouse hover on BEAUTY, FOOD... successful, dropdown visible.

☒ Mouse hover successfully performed on *Beauty, Food, Toys & More*. Dropdown submenu visible for 10 seconds.

Screenshot of Task 5 Execution:



Task 6 – Click on a Product

Goal: Click the first product, open it in a new tab, and switch using `getWindowHandles()`.

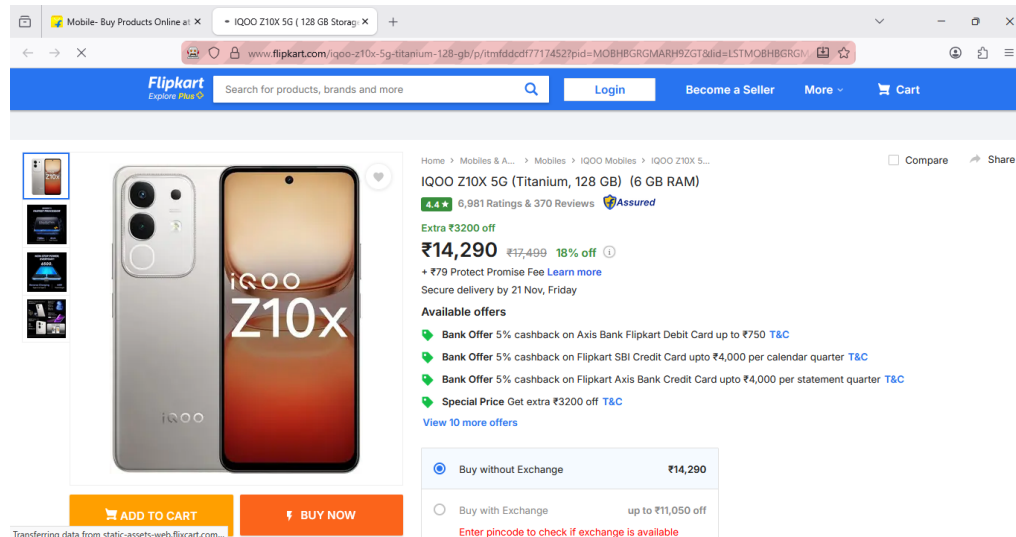
Output:

Opened product in a new tab: IQOO Z10X 5G (Titanium, 128 GB)

Switched to product details tab

Result:  Passed

Screenshot of Task 6 Execution:



Task 7 – Verify Product Details

Goal: Capture product title and price.

Output:

Product Name: IQOO Z10X 5G (Titanium, 128 GB) (6 GB RAM)

Product Price: ₹14,288

Result:  Passed

Screenshot of Task 7 Execution:

```
----- TASK 7: Verify Product Details -----  
Product Name: IQOO Z10X 5G (Titanium, 128 GB) (6 GB RAM)  
Product Price: ₹14,290
```

Task 8 – Take Screenshot

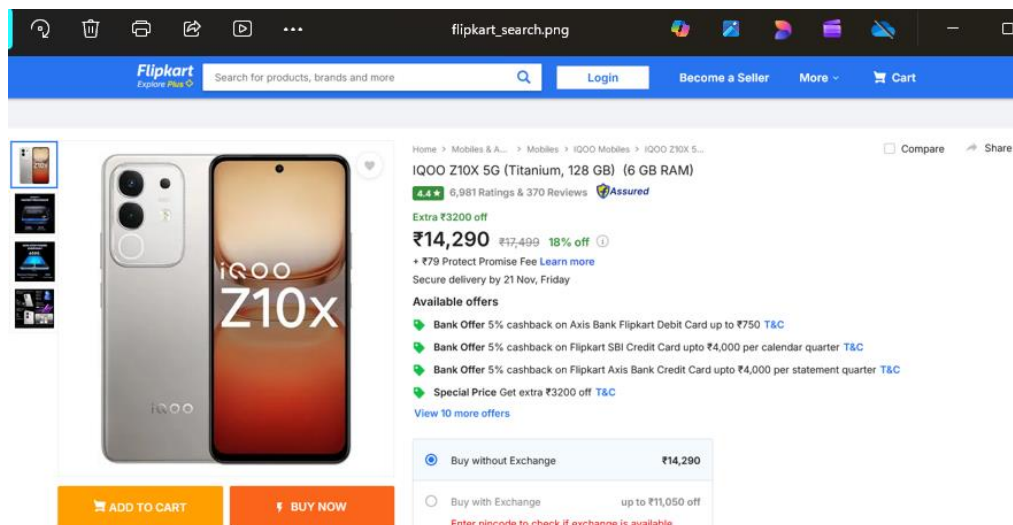
Goal: Capture the search results page and save it as *flipkart_search.png*.

Output:

Screenshot saved successfully!

Result:  Passed

Screenshot of Task 8 Execution:



Task 9 – Close Browser

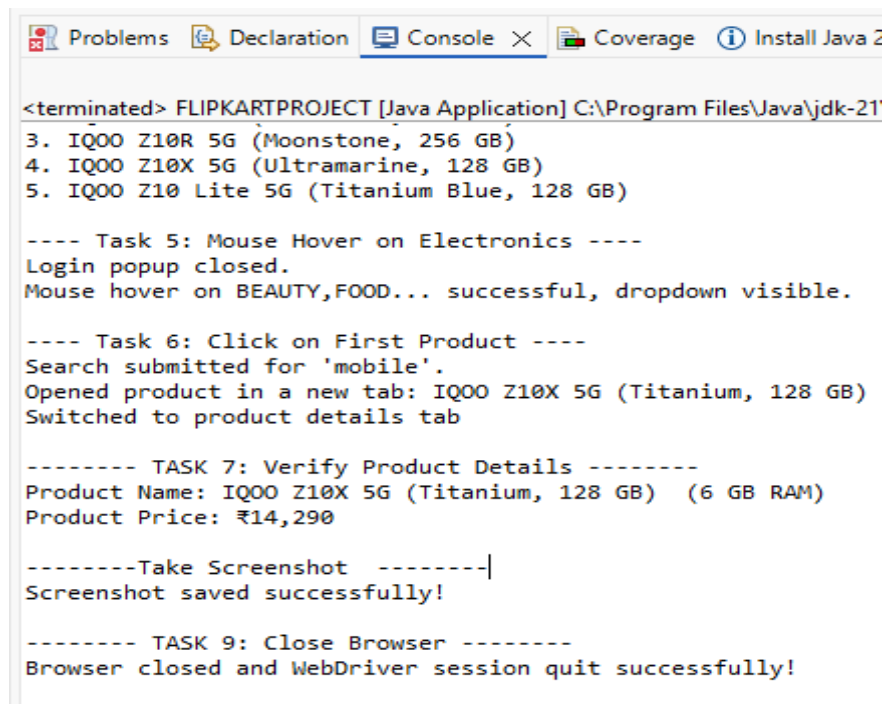
Goal: Close browser and end WebDriver session.

Output:

Browser closed and WebDriver session quit successfully!

Result: ☒ Passed

Screenshot of Task 9 Execution:



Test Case Summary

Task	Description	Result
1	Browser launch and title verification	✓ Passed
2	Handle login popup	✓ Passed
3	Search for product	✓ Passed
4	Capture product listings	✓ Passed
5	Mouse hover on menu	✓ Passed
6	Click product & handle new tab	✓ Passed
7	Verify product details	✓ Passed
8	Capture screenshot	✓ Passed
9	Close browser	✓ Passed

Total Executed: 9 Passed: 9 Failed: 0

Challenges and Solutions

Challenge	Reason	Solution
Dropdown not showing for Electronics/Fashion	Dynamic JS hover	Used Beauty & Food menu
Locators changed frequently	Dynamic React classes	Used XPath/text-based locators
Product opened in new tab	JavaScript execution using window.open()	Verified product opened successfully in a new browser tab
Dropdown vanished quickly	Hover timeout	Added 10-second pause
Login popup not always visible	Popup appears only on new browser session	Added conditional handling logic with try-catch

Learning Outcomes

- Understood how to handle dynamic web elements and pop-ups.
- Practiced multiple locator strategies and synchronization techniques.
- Gained hands-on experience with Actions class, JavaScriptExecutor, and window handling.

- Learned how to document automation projects professionally.

Conclusion

All core Flipkart functionalities were automated and verified successfully using Selenium WebDriver and Java.

Dynamic elements and pop-ups were handled effectively, and test documentation was completed in a professional format.

This project strengthened real-world automation testing and reporting skills.

Overall Result: ☒ Project Completed Successfully

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